ACE Model (Subset Data)

Step 0: Install and load required packages

```
# we recommend running this is a fresh R session or restarting your current session
install.packages(
    "cmdstanr",
    repos = c('https://stan-dev.r-universe.dev', getOption("repos"))
)
library(cmdstanr)
# Check toolchains
check_cmdstan_toolchain()
# If toolcahin is not setup properly, follow the printed instructions, or visit
https://mc-stan.org/docs/cmdstan-guide/installation.html#cpp-toolchain
# Install cmdstan
cmdstanr::install_cmdstan()
# We also install the bayesplot package visualizing MCMC results
install.packages("bayesplot")
```

```
library(cmdstanr)
```

```
This is cmdstanr version 0.9.0
```

```
- CmdStanR documentation and vignettes: mc-stan.org/cmdstanr
```

```
- CmdStan path: /home/marklai/.cmdstan/cmdstan-2.39.0
```

```
- CmdStan version: 2.39.0
```

```
register_knitr_engine()
options(mc.cores = min(parallel::detectCores() - 1, 4)) # Use four CPU cores for
MCMC sampling, if available
library(bayesplot)
```

```
This is bayesplot version 1.15.0
```

```
- Online documentation and vignettes at mc-stan.org/bayesplot
```

```
- bayesplot theme set to bayesplot::theme_default()
```

```
   * Does _not_ affect other ggplot2 plots
```

```
   * See ?bayesplot_theme_set for details on theme setting
```

```
library(HDInterval)
library(ggplot2)
```

```
# Define function for HDI
hdi2 <- function(x, cred_mass = 0.90) {
    hdi_interval <- HDInterval::hdi(x, credMass = cred_mass)
    setNames(hdi_interval, c("hdi_lo", "hdi_hi"))
}
```

Step 1: Data import

```
set.seed(1234)
n_mz <- c(325, 60, 7, 78, 150, 40, 8, 38, 88)
n_dz <- c(202, 97, 19, 96, 112, 64, 46, 56, 66)
# Convert to long format data frame
dat_mz <- data.frame(
    twin1 = rep(rep(1:3, 3), n_mz),
    twin2 = rep(rep(1:3, each = 3), n_mz)
)
dat_dz <- data.frame(
    twin1 = rep(rep(1:3, 3), n_dz),
    twin2 = rep(rep(1:3, each = 3), n_dz)
)
dat_mz[] <- lapply(dat_mz, ordered, labels = c("normal", "overweight", "obese"))
dat_dz[] <- lapply(dat_dz, ordered, labels = c("normal", "overweight", "obese"))
dat_full <- rbind(
    data.frame(zygosity = "MZ", dat_mz),
    data.frame(zygosity = "DZ", dat_dz)
)
sub_n <- 100
dat <- dat_full[sample.int(nrow(dat_full), 100), ]
# dat <- dat_full
```

```r
str(dat)
```

```
'data.frame':   100 obs. of  3 variables:
 $ zygosity: chr  "DZ" "DZ" "DZ" "DZ" ...
 $ twin1   : Ord.factor w/ 3 levels "normal"<"overweight"<..: 2 2 1 2 3 1 3 1
1 1 ...
 $ twin2   : Ord.factor w/ 3 levels "normal"<"overweight"<..: 2 1 2 1 2 1 2 1
2 1 ...
```

```r
summary(dat)
```

```
     zygosity          twin1           twin2
 Length   :100   normal    :53   normal    :48
 N.unique :  2   overweight:29   overweight:41
 N.blank  :  0   obese     :18   obese     :11
 Min.nchar:  2
 Max.nchar:  2
```

```r
tab_mz <- with(dat[dat$zygosity == "MZ", ], table(twin1, twin2))
tab_dz <- with(dat[dat$zygosity == "DZ", ], table(twin1, twin2))
```

Step 2: Compile the Stan model

```
//
// This Stan program estimates the polychoric correlation between two
// ordinal variables, and decompose based on an ACE model. This model
// assumes symmetry in the contingency table.
//
functions {
  #include "ace_fun.stan"
}
data {
  int<lower=0> k;  // number of categories
  array[k * (k + 1) %/% 2] int<lower=0> ltri_ns1;  // mz
  array[k * (k + 1) %/% 2] int<lower=0> ltri_ns2;  // dz
}
transformed data {
  int kprime = k * (k + 1) %/% 2;
  int N1 = sum(ltri_ns1);
  real N1real = N1;
  int N2 = sum(ltri_ns2);
```

```
  real N2real = N2;
}
parameters {
  real<lower=0, upper=1> scaled_rho1;
  real<lower=0, upper=1> scaled_rho2;
  ordered[k - 1] thres;
}
transformed parameters {
  real rho1 = scaled_rho1 * 2 - 1;
  real rho2 = scaled_rho2 * 2 - 1;
  vector<lower=0>[kprime] theta1 =
    vec_symm(polycor_to_prob(thres, rho1));
  vector<lower=0>[kprime] theta2 =
    vec_symm(polycor_to_prob(thres, rho2));
}
// The model to be estimated.
model {
  scaled_rho1 ~ beta(1.5, 1.5);
  scaled_rho2 ~ beta(1.5, 1.5);
  thres ~ student_t(4, 0, 2.5);
  ltri_ns1 ~ multinomial(theta1);
  ltri_ns2 ~ multinomial(theta2);
}
generated quantities {
  real h2 = 2 * (rho1 - rho2);
  real c2 = 2 * rho2 - rho1;
  real e2 = 1 - rho1;
  array[kprime] int ltri_ns1_tilde = multinomial_rng(theta1, N1);
  array[kprime] int ltri_ns2_tilde = multinomial_rng(theta2, N2);
  real X2_1_samp = 0.0;
  real X2_2_samp = 0.0;
  int<lower=0, upper=1> X2_1_gt;
  int<lower=0, upper=1> X2_2_gt;
  {
    vector[kprime] N1theta = N1real .* theta1;
    vector[kprime] N2theta = N2real .* theta2;
    real X2_1_rep = 0.0;
    real X2_2_rep = 0.0;
    for (idx in 1:kprime) {
      real ns1_exp = N1theta[idx];
      real ns2_exp = N2theta[idx];
      X2_1_samp += (ltri_ns1[idx] - ns1_exp)^2 / ns1_exp;
      X2_2_samp += (ltri_ns2[idx] - ns2_exp)^2 / ns2_exp;
```

```
      X2_1_rep += (ltri_ns1_tilde[idx] - ns1_exp)^2 / ns1_exp;
      X2_2_rep += (ltri_ns2_tilde[idx] - ns2_exp)^2 / ns2_exp;
    }
    X2_1_gt = X2_1_rep > X2_1_samp;
    X2_2_gt = X2_2_rep > X2_2_samp;
  }
}
```

Step 3: Prepare data and sampling

```
# Reduced frequency vector
to_ltri <- function(tab_x) {
    symm_x <- tab_x + t(tab_x) - diag(diag(tab_x))
    ltri_x <- symm_x[lower.tri(symm_x, diag = TRUE)]
}
(ltri_nmz <- to_ltri(tab_mz))
```

```
[1] 24  5  2 15  5  4
```

```
(ltri_ndz <- to_ltri(tab_dz))
```

```
[1] 14 12  6  6  6  1
```

```
ace_fit <- ace$sample(
    list(k = 3, ltri_ns1 = ltri_nmz, ltri_ns2 = ltri_ndz),
    chains = 4,
    iter_warmup = 1000,
    iter_sampling = 1000,
    seed = 732 # for reproducibility
)
```

Step 4: Check convergence

```
params <- c("rho1", "rho2", "thres", "h2", "c2", "e2")
ace_fit$print(params)
```

```
 variable  mean median   sd  mad    q5  q95 rhat ess_bulk ess_tail
 rho1      0.77   0.78 0.08 0.08  0.62 0.88 1.00     3651     3155
 rho2      0.17   0.18 0.17 0.18 -0.13 0.45 1.00     3444     2771
 thres[1]  0.02   0.03 0.10 0.10 -0.14 0.19 1.00     3946     2854
```

```
 thres[2]  1.05   1.05 0.12 0.12  0.85 1.26 1.00     4993     3488
 h2        1.19   1.18 0.38 0.38  0.58 1.83 1.00     3688     2911
 c2       -0.42  -0.41 0.36 0.36 -1.03 0.14 1.00     3559     2712
 e2        0.23   0.22 0.08 0.08  0.12 0.38 1.00     3651     3155
```

```
ace_fit$diagnostic_summary()
```

```
$num_divergent
[1] 0 0 0 0

$num_max_treedepth
[1] 0 0 0 0

$ebfmi
[1] 1.052552 1.072254 1.016314 1.094353
```

```
ace_fit$draws(params) |>
    mcmc_rank_hist()
```

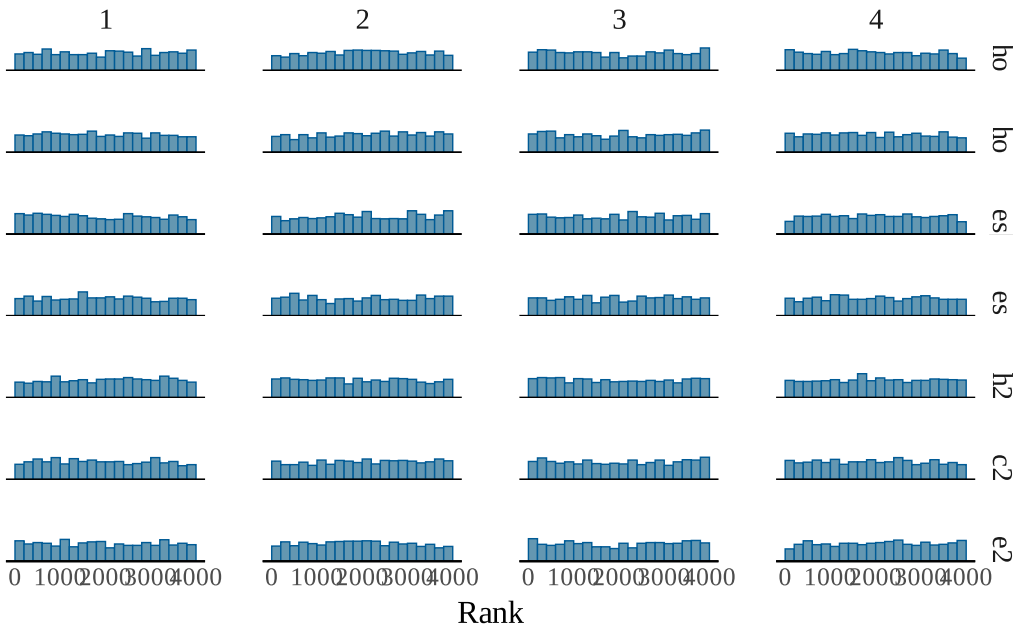


Figure 1: Rank histograms for the main model parameters

Step 5: Model checking

Repeat Steps 3 and 4 with reparameterized model

With unreasonable h2 and c2, we reparameterize the model to enforce theory-informed priors/ constraints.

```
//
// This Stan program estimates the polychoric correlation between two
// ordinal variables, and decompose based on an ACE model. The model is
// parameterized using a simplex for the ACE components to enforce
// nonzero variances, and assumes symmetry in the contingency table.
//
functions {
  #include "ace_fun.stan"
}
data {
  int<lower=0> k;  // number of categories
  array[k * (k + 1) %/% 2] int<lower=0> ltri_ns1;  // mz
  array[k * (k + 1) %/% 2] int<lower=0> ltri_ns2;  // dz
}
transformed data {
  int kprime = k * (k + 1) %/% 2;
  int N1 = sum(ltri_ns1);
  real N1real = N1;
  int N2 = sum(ltri_ns2);
  real N2real = N2;
}
parameters {
  simplex[3] ace;
  ordered[k - 1] thres;
}
transformed parameters {
  real rho1 = ace[1] + ace[2];
  real rho2 = ace[1] / 2 + ace[2];
  vector<lower=0>[kprime] theta1 =
    vec_symm(polycor_to_prob(thres, rho1));
  vector<lower=0>[kprime] theta2 =
    vec_symm(polycor_to_prob(thres, rho2));
}
// The model to be estimated.
model {
  thres ~ student_t(4, 0, 2.5);
  ltri_ns1 ~ multinomial(theta1);
  ltri_ns2 ~ multinomial(theta2);
```

```
}
generated quantities {
  real h2 = ace[1];
  real c2 = ace[2];
  real e2 = ace[3];
  array[kprime] int ltri_ns1_tilde = multinomial_rng(theta1, N1);
  array[kprime] int ltri_ns2_tilde = multinomial_rng(theta2, N2);
  real X2_1_samp = 0.0;
  real X2_2_samp = 0.0;
  int<lower=0, upper=1> X2_1_gt;
  int<lower=0, upper=1> X2_2_gt;
  {
    vector[kprime] N1theta = N1real .* theta1;
    vector[kprime] N2theta = N2real .* theta2;
    real X2_1_rep = 0.0;
    real X2_2_rep = 0.0;
    for (idx in 1:kprime) {
      real ns1_exp = N1theta[idx];
      real ns2_exp = N2theta[idx];
      X2_1_samp += (ltri_ns1[idx] - ns1_exp)^2 / ns1_exp;
      X2_2_samp += (ltri_ns2[idx] - ns2_exp)^2 / ns2_exp;
      X2_1_rep += (ltri_ns1_tilde[idx] - ns1_exp)^2 / ns1_exp;
      X2_2_rep += (ltri_ns2_tilde[idx] - ns2_exp)^2 / ns2_exp;
    }
    X2_1_gt = X2_1_rep > X2_1_samp;
    X2_2_gt = X2_2_rep > X2_2_samp;
  }
}
```

One can then repeat steps 3 and 4 to obtain the posterior samples and convergence diagnostics, which we do not repeat the details here. The posterior summaries of the main parameters are in Table 1.

```
ace_s_fit <- ace_simplex$sample(
    list(k = 3, ltri_ns1 = ltri_nmz, ltri_ns2 = ltri_ndz),
    chains = 4,
    iter_warmup = 1000,
    iter_sampling = 1000,
    seed = 732 # for reproducibility
)
```

```
ace_s_fit$summary(params, mean, median, sd, mad, posterior::quantile2, hdi2) |>
    transform(
        variable = paste0(
            variable,
            " (",
            c(
                "$\\rho^\\text{MZ}$",
                "$\\rho^\\text{DZ}$",
                "$\\tau_1$",
                "$\\tau_2$",
                "$h^2$",
                "$c^2$",
                "$e^2$"
            ),
            ")"
        )
    ) |>
    knitr::kable(digits = 2)
```

Table 1: Posterior summary of the main model parameters, based on the simplex parameterization

variable	mean	median	sd	mad	q5	q95	hdi_lo	hdi_hi
rho1 (ρ^{MZ})	0.75	0.76	0.09	0.09	0.58	0.87	0.61	0.89
rho2 (ρ^{DZ})	0.45	0.44	0.07	0.07	0.34	0.58	0.33	0.57
thres[1] (τ_1)	0.03	0.03	0.10	0.11	−0.14	0.20	−0.14	0.19
thres[2] (τ_2)	1.04	1.04	0.13	0.13	0.83	1.25	0.82	1.23
h2 (h^2)	0.60	0.63	0.16	0.16	0.29	0.81	0.35	0.84
c2 (c^2)	0.15	0.12	0.12	0.11	0.01	0.39	0.00	0.32
e2 (e^2)	0.25	0.24	0.09	0.09	0.13	0.42	0.11	0.39

Global fit

Local fit

```
x_scale <- scale_x_continuous(
    breaks = seq_along(ltri_nmz),
    labels = c(
        "(1,1)",
        "(2,1)+(1,2)",
        "(2,2)",
```

```
        "(3,1)+(1,3)",
        "(3,2)+(2,3)",
        "(3,3)"
    )
)
```

```
ace_s_fit$draws("ltri_ns1_tilde", format = "matrix") |>
    ppc_intervals(y = ltri_nmz) +
    xlab("Cell (twin1, twin2)") +
    x_scale
```

```
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
ℹ Please use `linewidth` instead.
ℹ The deprecated feature was likely used in the bayesplot package.
  Please report the issue at <https://github.com/stan-dev/bayesplot/issues/>.
```

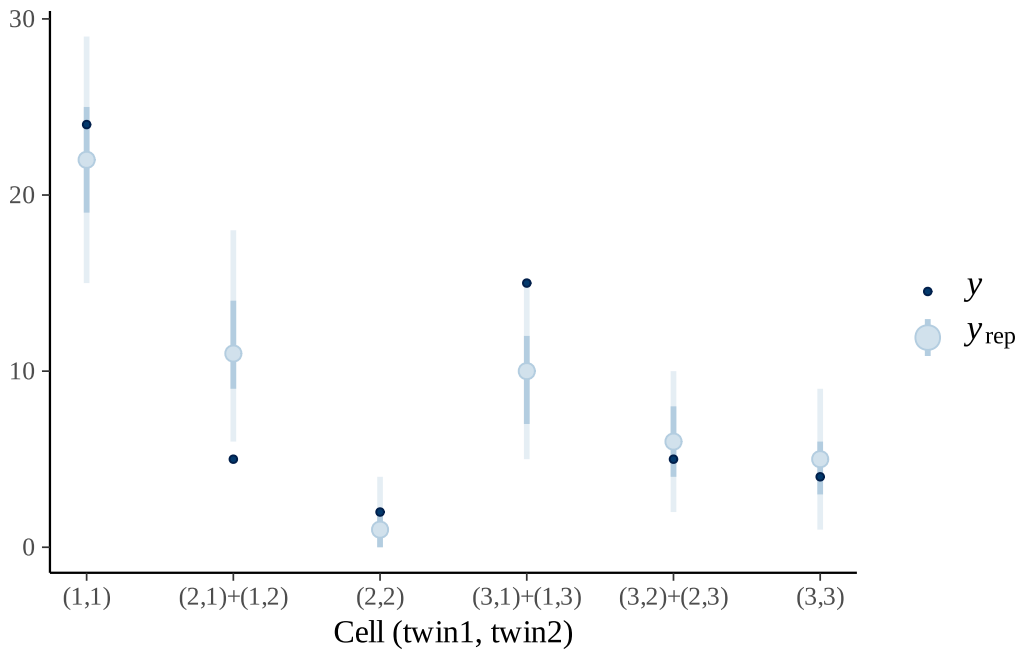


Figure 2: Posterior predictive check for each cell count in the contingency tables for MZ twins

```
ace_s_fit$draws("ltri_ns2_tilde", format = "matrix") |>
    ppc_intervals(y = ltri_ndz) +
    xlab("Cell (twin1, twin2)") +
    x_scale
```

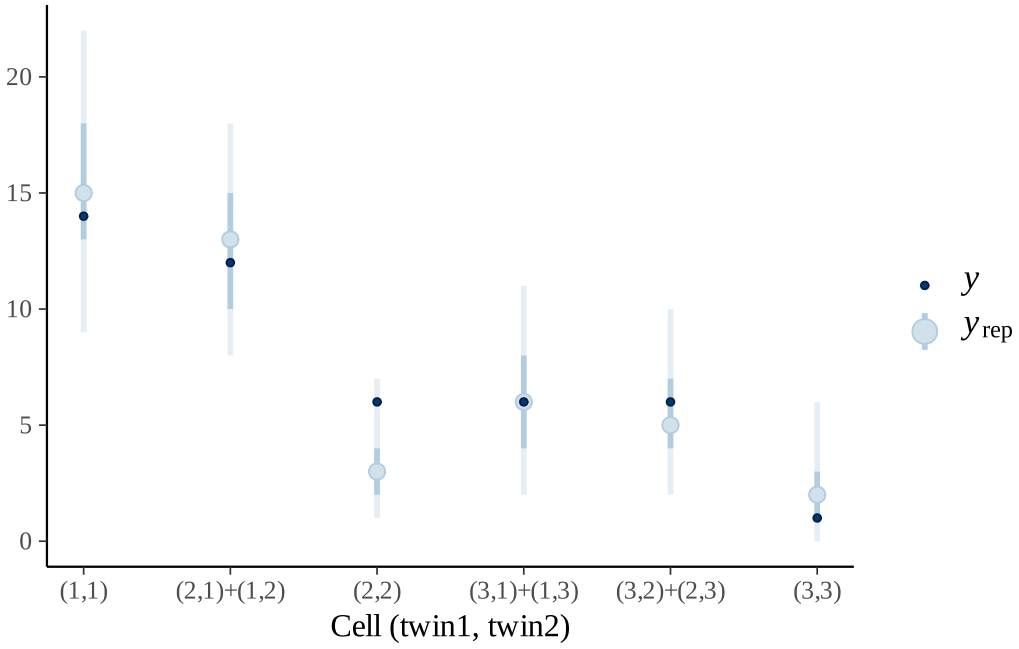


Figure 3: Posterior predictive check for each cell count in the contingency tables for DZ twins

Step 6: Summarizing the posterior

```
ace_s_fit$draws(params) |>
    mcmc_hist()
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

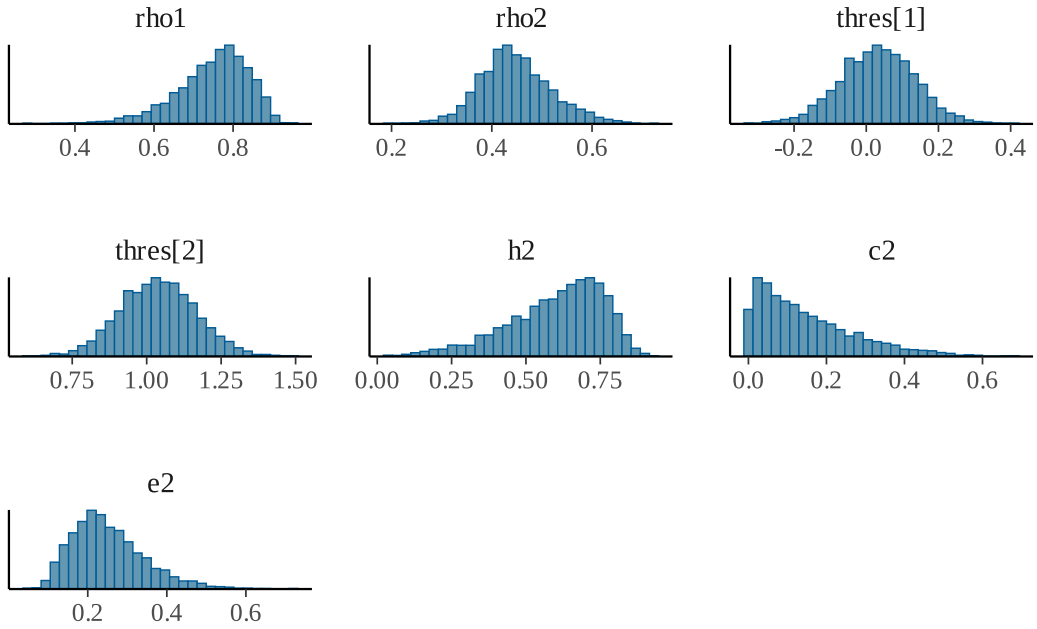


Figure 4: Posterior distributions of the main model parameters

Compare with *umx* package

```
library(umx)
```

```
Loading required package: OpenMx
```

```
To take full advantage of multiple cores, use:
  mxOption(key='Number of Threads', value=parallel::detectCores()) #now
  Sys.setenv(OMP_NUM_THREADS=parallel::detectCores()) #before library(OpenMx)
```

```
For an overview type '?umx'
```

```

Attaching package: 'umx'
```

```
The following object is masked from 'package:stats':

    loadings
```

```r
# Make the ordinal variables into umxFactors
dat[, c("twin1", "twin2")] <- umxFactor(dat[, c("twin1", "twin2")])

mz_data <- dat[dat$zygosity == "MZ", ]
dz_data <- dat[dat$zygosity == "DZ", ]

# Model and summary with FIML
m1 <- umxACE(
    selDVs = "twin",
    dzData = dz_data,
    mzData = mz_data,
    sep = "",
    tryHard = "ordinal"
)
```

```
Found 1 pair(s) of ordinal variables:'twin1' and 'twin2' (No binary)
```

```
Running ACE with 4 parameters
```

```

Beginning initial fit attempt
```

```
Running ACE with 4 parameters
```

```
 Lowest minimum so far:  365.978829144973
```

```

Beginning fit attempt 1 of at maximum 10 extra tries
```

```
Running ACE with 4 parameters
```

```

Beginning fit attempt 2 of at maximum 10 extra tries
```

```
Running ACE with 4 parameters
```

```
 Lowest minimum so far:  365.978829144971
```

```
Beginning fit attempt 3 of at maximum 10 extra tries
```

```
Running ACE with 4 parameters
```

```
 Lowest minimum so far:  365.978829144971
```

```
Beginning fit attempt 4 of at maximum 10 extra tries
```

```
Running ACE with 4 parameters
```

```
 Lowest minimum so far:  365.97882914497
```

```
Beginning fit attempt 5 of at maximum 10 extra tries
```

```
Running ACE with 4 parameters
```

```
Beginning fit attempt 6 of at maximum 10 extra tries
```

```
Running ACE with 4 parameters
```

```
Beginning fit attempt 7 of at maximum 10 extra tries
```

```
Running ACE with 4 parameters
```

```
Beginning fit attempt 8 of at maximum 10 extra tries
```

```
Running ACE with 4 parameters
```

```
Beginning fit attempt 9 of at maximum 10 extra tries
```

```
Running ACE with 4 parameters
```

```
Beginning fit attempt 10 of at maximum 10 extra tries
```

```
Running ACE with 4 parameters
```

```
Retry limit reached
```

```
Solution found
```

```
Final run, for Hessian and/or standard errors and/or confidence intervals
```

```
Running ACE with 4 parameters
```

```
 Solution found!  Final fit=365.97883 (started at 370.92682)  (11 attempt(s):
11 valid, 0 errors)
```

```
 Start values from best fit:
```

```
-0.024758927371081,0.934118071479837,0,0.471878643174057
```

```
ACE -2 × log(Likelihood) = 365.979
```

```
Table: Standardized parameter estimates from a 1-factor Cholesky ACE model. A:
additive genetic; C: common environment; E: unique environment.

|     |    a1|c1 |    e1|
|:----|-----:|:--|-----:|
|twin | 0.893|.  | 0.451|


Table: Means (from model$top$expMean)

|          |  twin1|  twin2|
|:---------|------:|------:|
|intercept | -0.025| -0.025|
```

```
?umxPlotACE options: std=T/F, means=T/F, digits=n, strip_zero=T/F, file=, min=,
max =
```

```
umxSummary(m1)
```

```
ACE -2 × log(Likelihood) = 365.98
```

```
Table: Standardized parameter estimates from a 1-factor Cholesky ACE model. A:
additive genetic; C: common environment; E: unique environment.

|     |   a1|c1 |   e1|
|:----|----:|:--|----:|
|twin | 0.89|.  | 0.45|


Table: Means (from model$top$expMean)

|          |  twin1|  twin2|
|:---------|------:|------:|
|intercept | -0.025| -0.025|
```

```
?umxPlotACE options: std=T/F, means=T/F, digits=n, strip_zero=T/F, file=, min=, max =
```

```
# CI failed
umxConfint(m1, run = TRUE)
```

```
Running ACE with 4 parameters
```

```
               lbound estimate ubound lbound Code ubound Code
top.a_std[1,1]     NA    0.893     NA          NA          NA
top.c_std[1,1]     NA    0.000  0.577          NA           3
top.e_std[1,1]     NA    0.451     NA          NA          NA
```

```
# Model and summary with DWLS
m1_dwls <- umxACE(
    selDVs = "twin",
    dzData = dz_data,
    mzData = mz_data,
    sep = "",
    type = "DWLS",
    tryHard = "ordinal"
)
```

```
Found 1 pair(s) of ordinal variables:'twin1' and 'twin2' (No binary)
```

```
Data treated as DWLS
```

```
Running ACE with 4 parameters
```

```
Beginning initial fit attempt
```

```
Running ACE with 4 parameters
```

```
 Lowest minimum so far:  0.0391834784698901
```

```
Beginning fit attempt 1 of at maximum 10 extra tries
```

```
Running ACE with 4 parameters
```

```
 Lowest minimum so far:  0.0391834784698893
```

```
Beginning fit attempt 2 of at maximum 10 extra tries
```

```
Running ACE with 4 parameters
```

```
 Lowest minimum so far:  0.0391834784698892
```

```
Beginning fit attempt 3 of at maximum 10 extra tries
```

```
Running ACE with 4 parameters
```

```
Beginning fit attempt 4 of at maximum 10 extra tries
```

```
Running ACE with 4 parameters
```

```
Beginning fit attempt 5 of at maximum 10 extra tries
```

```
Running ACE with 4 parameters
```

```
 Lowest minimum so far:  0.0391834784698892
```

```
Beginning fit attempt 6 of at maximum 10 extra tries
```

```
Running ACE with 4 parameters
```

```
Beginning fit attempt 7 of at maximum 10 extra tries
```

```
Running ACE with 4 parameters
```

```
Beginning fit attempt 8 of at maximum 10 extra tries
```

```
Running ACE with 4 parameters
```

```
 Lowest minimum so far:  0.0391834784698892
```

```
Beginning fit attempt 9 of at maximum 10 extra tries
```

```
Running ACE with 4 parameters
```

```
Beginning fit attempt 10 of at maximum 10 extra tries
```

```
Running ACE with 4 parameters
```

```
Retry limit reached
```

```
Solution found
```

```
Final run, for Hessian and/or standard errors and/or confidence intervals
```

```
Running ACE with 4 parameters
```

```
 Solution found!  Final fit=0.039183478 (started at 0.10351156)  (11 attempt(s):
11 valid, 0 errors)
```

```
 Start values from best fit:
```

```
-8.39772023278525e-05,0.895116918606286,9.4084489021653e-09,0.461643857432493
```

```
ACE -2 × log(Likelihood) = 6.478
```

```
Table: Standardized parameter estimates from a 1-factor Cholesky ACE model. A:
additive genetic; C: common environment; E: unique environment.

|     |    a1|c1 |    e1|
|:----|-----:|:--|-----:|
|twin | 0.889|.  | 0.458|


Table: Means (from model$top$expMean)

|          |twin1 |twin2 |
|:---------|:-----|:-----|
|intercept |0     |0     |
```

```
?umxPlotACE options: std=T/F, means=T/F, digits=n, strip_zero=T/F, file=, min=,
max =
```

```
umxSummary(m1_dwls)
```

```
ACE -2 × log(Likelihood) = 6.48
```

```
```

```
Table: Standardized parameter estimates from a 1-factor Cholesky ACE model. A:
additive genetic; C: common environment; E: unique environment.

|     |   a1|c1 |   e1|
|:----|----:|:--|----:|
|twin | 0.89|.  | 0.46|


Table: Means (from model$top$expMean)

|          |twin1 |twin2 |
|:---------|:-----|:-----|
|intercept |0     |0     |
```

```
?umxPlotACE options: std=T/F, means=T/F, digits=n, strip_zero=T/F, file=, min=,
max =
```

```
# Extremely large standard errors
summary(m1_dwls)
```

```
Summary of ACE

free parameters:
           name      matrix   row   col      Estimate    Std.Error A lbound
1 expMean_twin1 top.expMean means twin1 -8.397720e-05 1.974439e+00
2        a_r1c1       top.a     1     1  8.951169e-01 1.039425e+00        0
3        c_r1c1       top.c     1     1  9.408449e-09 1.122410e-08 !     0!
4        e_r1c1       top.e     1     1  4.616439e-01 8.338411e-01        0
  ubound
1
2
3
4

To obtain confidence intervals re-run with intervals=TRUE

Model Statistics:
               |  Parameters  |  Degrees of Freedom  |  Fit (r'wr units)
       Model:              4                      6                   NA
   Saturated:             NA                     NA                   NA
```

```
Independence:            NA                    NA                    NA
Number of observations/statistics: 100/10

chi-square:  χ² ( df=NA ) = 6.477706,  p = 0.3718542
CFI: NA
TLI: NA   (also known as NNFI)
RMSEA:  0.02821661  [95% CI (0, 0.1500864)]
Prob(RMSEA <= 0.05): 0.5253774
To get additional fit indices, see help(mxRefModels)
timestamp: 2026-06-29 22:25:44
Wall clock time: 0.1229384 secs
optimizer:  SLSQP
OpenMx version number: 2.22.11
Need help?  See help(mxSummary)
```

```
# Model and summary with WLS
m1_wls <- umxACE(
    selDVs = "twin",
    dzData = dz_data,
    mzData = mz_data,
    sep = "",
    type = "WLS",
    tryHard = "ordinal"
)
```

```
Found 1 pair(s) of ordinal variables:'twin1' and 'twin2' (No binary)
```

```
Data treated as WLS
```

```
Running ACE with 4 parameters
```

```

Beginning initial fit attempt
```

```
Running ACE with 4 parameters
```

```

 Lowest minimum so far:  0.078817106554107
```

```
Beginning fit attempt 1 of at maximum 10 extra tries
```

```
Running ACE with 4 parameters
```

```
Beginning fit attempt 2 of at maximum 10 extra tries
```

```
Running ACE with 4 parameters
```

```
Beginning fit attempt 3 of at maximum 10 extra tries
```

```
Running ACE with 4 parameters
```

```
 Lowest minimum so far:  0.0788171065541069
```

```
Beginning fit attempt 4 of at maximum 10 extra tries
```

```
Running ACE with 4 parameters
```

```
Beginning fit attempt 5 of at maximum 10 extra tries
```

```
Running ACE with 4 parameters
```

```
 Lowest minimum so far:  0.0788171065541069
```

```
Beginning fit attempt 6 of at maximum 10 extra tries
```

```
Running ACE with 4 parameters
```

```
Beginning fit attempt 7 of at maximum 10 extra tries
```

```
Running ACE with 4 parameters
```

```
Beginning fit attempt 8 of at maximum 10 extra tries
```

```
Running ACE with 4 parameters
```

```
Beginning fit attempt 9 of at maximum 10 extra tries
```

```
Running ACE with 4 parameters
```

```
Beginning fit attempt 10 of at maximum 10 extra tries
```

```
Running ACE with 4 parameters
```

```
Retry limit reached
```

```
Solution found
```

```
Final run, for Hessian and/or standard errors and/or confidence intervals
```

```
Running ACE with 4 parameters
```

```
 Solution found!  Final fit=0.078817107 (started at 0.20654351)  (11 attempt(s):
11 valid, 0 errors)
```

```
 Start values from best fit:
```

```
-0.0342164605804386,0.971005705861823,0,0.447841878956487
```

```
ACE -2 × log(Likelihood) = 7.882
```

```
Table: Standardized parameter estimates from a 1-factor Cholesky ACE model. A:
additive genetic; C: common environment; E: unique environment.

|     |    a1|c1 |    e1|
|:----|-----:|:--|-----:|
|twin | 0.908|.  | 0.419|


Table: Means (from model$top$expMean)

|          |  twin1|  twin2|
|:---------|------:|------:|
|intercept | -0.034| -0.034|
```

```
?umxPlotACE options: std=T/F, means=T/F, digits=n, strip_zero=T/F, file=, min=,
max =
```

```
umxSummary(m1_wls)
```

```
ACE -2 × log(Likelihood) = 7.88
```

```
Table: Standardized parameter estimates from a 1-factor Cholesky ACE model. A:
additive genetic; C: common environment; E: unique environment.

|     |   a1|c1 |   e1|
|:----|----:|:--|----:|
|twin | 0.91|.  | 0.42|


Table: Means (from model$top$expMean)
```

```
|          |  twin1|  twin2|
|:---------|------:|------:|
|intercept | -0.034| -0.034|
```

```
?umxPlotACE options: std=T/F, means=T/F, digits=n, strip_zero=T/F, file=, min=,
max =
```

```
# CI failed
umxConfint(m1_wls, run = TRUE)
```

```
Running ACE with 4 parameters
```

```
               lbound estimate ubound lbound Code ubound Code
top.a_std[1,1]     NA    0.908     NA          NA          NA
top.c_std[1,1]     NA    0.000     NA          NA          NA
top.e_std[1,1]     NA    0.419     NA          NA          NA
```